

Assignment 1: Imitation Learning

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Collaborators: OpenAI Codex (For code debugging and plotting)

1 Behavioral Cloning (10 pt)

1.1 Part 2 (1.5 pt)

Table 1: Mean and standard deviation of expert return over two trajectories, for each environment.

Metric/Env	Ant-v2	Humanoid-v2	Walker2d-v2	Hopper-v2	HalfCheetah-v2
Mean	4713.65	10344.52	5566.85	3772.67	4205.78
Std.	12.20	20.98	9.24	1.95	83.04

1.2 Part 3 (5 pt)

Table 2: BC vs. expert return. List the environment you chose and the hyperparameters (network size, amount of data, training iterations, `eval_batch_size`, etc.) here.

Env	Ant-v2		Humanoid	
Metric	Mean	Std.	Mean	Std.
Expert	4713.65	12.20	10344.52	20.98
BC	4591.48	55.24	359.62	96.22

1.3 Part 4 (3 pt)

Ant-v2: Effect of Expert Data Size on Behavior Cloning

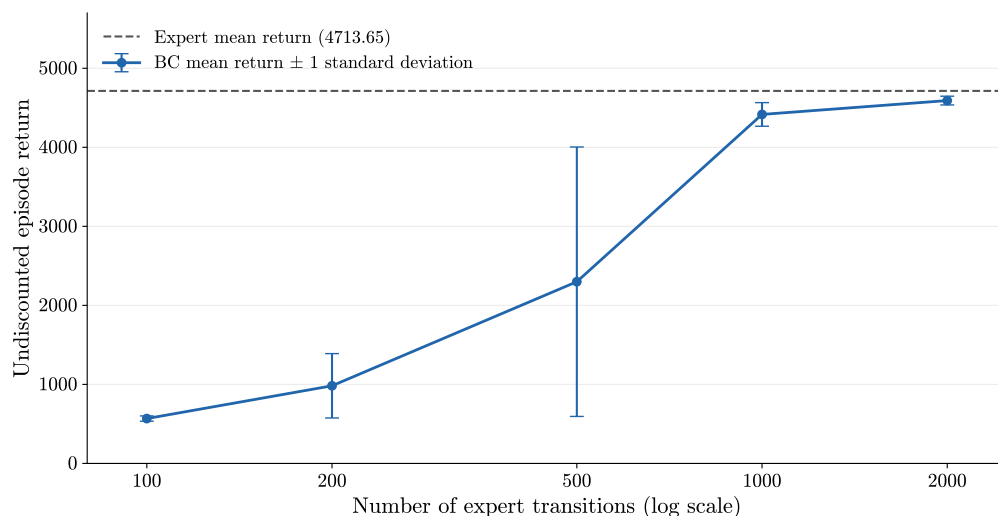


Figure 1: Effect of expert-data size on behavior cloning in Ant-v2. We hypothesized that reducing expert data would lower performance through reduced state-action coverage. Training used nested random subsets of 100, 200, 500, 1000, and 2000 expert transitions. Each policy used five hidden layers of 64 units, a learning rate of 0.004, and 5000 gradient updates with minibatches of 100. These hyperparameters were retained from our successful full-data Ant baseline and held fixed to isolate the effect of expert-data size.

1.4 Part 5 (0.5 pt)

Find one robotics paper from the last ~ 3 years that uses imitation learning (i.e. behavior cloning) as the primary methodology. In a short paragraph (4-5 sentences): (i) cite it, (ii) say what robot task is being cloned and where the expert data comes from and (iii) name a main IL risk from this homework and whether that still applies to the paper's methodology. A few sentences is enough; this is not meant to be an extensive literature review.

Chen et al. [1] use an imitation-based Diffusion Policy as the action policy for a real-world robotic manipulation system. The robot performs pick-and-place tasks in a toy kitchen, and the policy is trained from 120 teleoperated human demonstrations using wrist and third-person camera observations. A main risk of behavior cloning is distribution shift: once the learned policy encounters observations outside the expert demonstration distribution, its prediction errors can compound and lead it into increasingly unfamiliar states. This risk still applies here, and is particularly relevant because the paper studies novel visual distractors at test time that are outside the training distribution.

References

- [1] Yuxin Chen, Jianglan Wei, Chenfeng Xu, Boyi Li, Masayoshi Tomizuka, Andrea Bajcsy, and Ran Tian. Reimagination with test-time observation interventions: Distractor-robust world model predictions for visual model predictive control. In *Proceedings of the IEEE International Conference on Robotics and Automation*, 2026.

2 DAgger (5 pt)

2.1 Part 2 (5 pt)

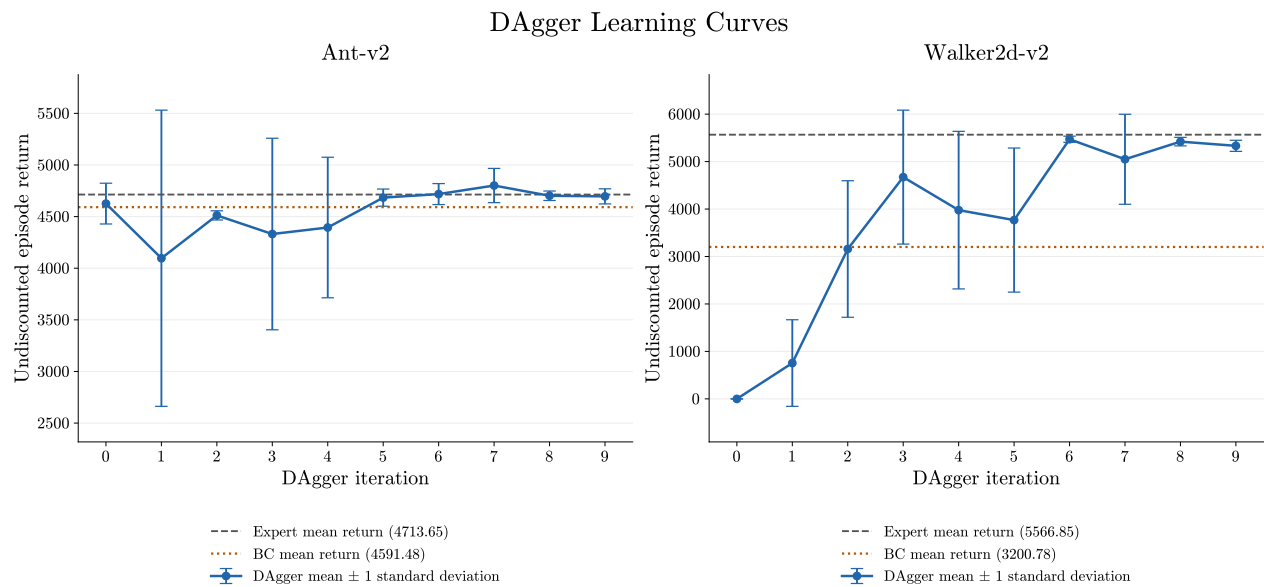


Figure 2: DAgger learning curves for Ant-v2 (left) and Walker2d-v2 (right). Each policy used five hidden layers of 64 units, a learning rate of 0.004, and 1000 gradient updates per iteration with minibatches of 100. Training began with 2000 expert transitions, followed by nine iterations that each collected at least 1000 additional transitions using the current policy, relabeled them with expert actions, and trained on the accumulated dataset. Evaluation collected at least 5000 steps per iteration, with a maximum episode length of 1000.